

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

TRAINING TR-102 REPORT DAY 23

23 JULY 2025

Overview:

Today, I successfully developed and tested a complete real-time face recognition-based attendance system. This system uses a webcam to capture live video, detects and recognizes faces by comparing them to known encodings, and logs the attendance of recognized individuals into a CSV file with the exact timestamp. The complete solution was built using Python and the face recognition and OpenCV libraries, without relying on Haar cascades, focusing solely on deep-learning-based face encodings.

Learning Objectives:

- Understand how to compare live facial input with pre-stored known encodings.
- Implement real-time face recognition using `face_recognition` and OpenCV.
- Automate attendance logging using pandas and CSV file handling.
- Distinguish known vs. unknown faces and handle each case effectively.
- Integrate all components into a fully functional and user-friendly project.

What I Implemented Today:

1. Face Encoding Setup:
 - a. Loaded and encoded images from a dataset/ folder using `face_recognition`.
 - b. Created a list of names associated with each encoding for future comparison.
2. Attendance Logging Logic:
 - a. Used the panda's library to create or update a CSV file `attendance.csv`.
 - b. Logged names and current timestamps only if the individual was not previously marked present.
3. Live Recognition with OpenCV:
 - a. Captured video from the webcam using `cv2.VideoCapture(0)`.
 - b. Detected and encoded faces in each frame and compared them using:

- i. `compare_faces` to check for matches.
 - ii. `face_distance` to determine the closest known face.
 - c. Drew bounding boxes (green for known faces, red for unknown) with names using `cv2.rectangle` and `cv2.putText`.
4. Handling Unknown Faces:
- a. Implemented a fallback to label unrecognized faces as “Unknown” and display them with a red box.
 - b. Ensured the program continues smoothly without crashing.

Real-Time Face Recognition Workflow

It focused on building a fully functional real-time face recognition system using Python. I implemented the project using the `face_recognition` library, which uses 128-dimensional face encodings for high-accuracy comparison. The system detects faces from a live webcam feed, compares them with pre-encoded known faces, and distinguishes between recognized and unrecognized individuals. I also integrated a live attendance system that logs the name and timestamp into a CSV file whenever a known face is identified. The interface was designed using OpenCV, allowing us to draw real-time bounding boxes and labels, making the user experience intuitive and informative.

Conclusion:

This final implementation marks the successful completion of the project, bringing all learned concepts together. I have now built a fully working face recognition system from scratch that can:

- Recognize multiple known faces in real-time,
- Log their attendance accurately,
- Handle unknown entries safely and clearly,
- And do so efficiently using lightweight tools like `face_recognition`, OpenCV, and pandas.

This project helped me gain in-depth experience with real-world AI/ML integration, file handling, and computer vision techniques. I now feel more confident about applying machine learning in hands-on applications like surveillance, smart attendance, and personalized recognition systems.

